AP Precalculus Unit 2 Practice Test

Exponential and Logarithmic Functions

Name: _____ **Date:** _____

About 45 minutes. No calculator unless marked. Show your work.

1. (2.1–2.2) Linear, exponential or neither?

Decide for each table. Justify using differences or ratios.

x	0	1	2	3	4
(a) $f(x)$	2	5	8	11	14
(b) $g(x)$	3	6	12	24	48
(c) $h(x)$	1	4	9	16	25

(d) Write a formula for the one that is exponential.

2. (2.3) Exponential function features

$f(x) = 200(0.85)^x$

(a) Is this growth or decay? By what percent does f change each time x increases by 1?

(b) Find $f(3)$. *(Calculator OK)*

(c) Describe the end behavior: $\lim_{x\to\infty} f(x)$ and $\lim_{x\to-\infty} f(x)$.

(d) What is the horizontal asymptote?

3. (2.4) Exponential model in context

A city has 50,000 people and its population **decreases by 3% each year**.

(a) Write a model $P(t)$ for the population after t years.

(b) Find the population after 10 years. *(Calculator OK)*

(c) Use logarithms to find when the population will be half of its original size. *(Calculator OK)*

4. (2.5) Choosing a model

Two models were fit to the same 7 data points. Their residuals are below.

x	1	2	3	4	5	6	7
Model A (linear) residual	5	1	-2	-3	-2	1	5
Model B (exponential) residual	1	-2	1	2	-1	-2	1

(a) Which model is a better fit? Explain using the *pattern* of the residuals.

(b) What does a residual of -3 mean?

5. (2.6) Composition

Let $f(x) = 2x + 1$ and $g(x) = x^2 - 3$.

(a) Find $f(g(2))$.

(b) Find $g(f(x))$ and simplify.

(c) Is $f(g(x)) = g(f(x))$? Explain.

6. (2.7) Inverse functions

(a) Find $f^{-1}(x)$ for $f(x) = 3x - 6$.

(b) A one-to-one function has $f(2) = 7$ and $f(5) = -1$. Find $f^{-1}(7)$ and $f^{-1}(-1)$.

(c) Does $h(x) = x^2$ (all real numbers) have an inverse function? Explain.

7. (2.8) Evaluate without a calculator

(a) $\log_2 32$ (b) $\log_5 \dfrac{1}{25}$ (c) $\log_{1/2} 8$ (d) $\ln e^4$ (e) $\log 0.001$

8. (2.10) A logarithmic function

$g(x) = \log_3(x-2) + 1$

(a) State the domain and the vertical asymptote.

(b) Find $g(5)$.

(c) Find the x-intercept.

9. (2.11) Properties of logarithms

(a) Expand: $\log_2\left(\dfrac{8x^3}{y}\right)$

(b) Write as a single logarithm: $3\log x - 2\log y + \log 4$

(c) True or false: $\log(a+b) = \log a + \log b$. Explain.

10. (2.12) Solving equations

Solve each. Give exact answers; also a decimal for (b) and (d). *(Calculator OK for decimals only)*

(a) $3^{2x-1} = 81$

(b) $7^x = 20$

(c) $\log_2 x + \log_2(x-2) = 3$

(d) $\ln(x+1) = 2$

11. (2.13–2.14) Half-life and semi-log

A sample has 80 grams of a substance with a **half-life of 6 days**.

(a) Write a model $A(t)$ for the mass after t days.

(b) How much remains after 15 days? *(Calculator OK)*

(c) When will 5 grams remain?

(d) A semi-log plot shows $\log_2 A$ versus t. What kind of graph is it, and what is its slope?